Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Core Pure 2 – Further Complex Numbers

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. State de Moivre's theorem. **[2]**

2. Use de Moivre to find $\left(\cos\theta + i\sin\theta\right)^5$. **[3]**

3. Express $\cos 5\theta$ in terms of powers of $\cos\theta$. **[4]**

4. Express $\sin 3\theta$ in terms of powers of $\sin\theta$. **[4]**

5. Find the cube roots of unity. **[4]**

6. Find the square roots of i. **[4]**

7. Find $z^6 = 1$ and plot the roots on an Argand diagram. **[4]**

8. Express $\cos\theta + i\sin\theta$ in exponential form. **[2]**

9. Find $\left(1+i\right)^{10}$ using de Moivre. **[4]**

10. Show that $\cos 2\theta = 2\cos^2\theta - 1$ using de Moivre. **[3]**

11. Find the fifth roots of 32. **[4]**

12. Solve $z^4 + 4 = 0$. **[5]**

13. Find $\sum_{k=0}^{n-1} \cos \frac{2k\pi}{n}$. **[3]**

14. Express $z^n + \frac{1}{z^n}$ in terms of $\cos\theta$ if $z = \operatorname{cis}\theta$. **[3]**

15. Find the real and imaginary parts of $e^{i\pi/3}$. **[2]**

Section B – Hard (Q16–25)

16. Express $\cos 6\theta$ in terms of $\cos\theta$. **[5]**

17. Find all solutions of $z^5 = 1 + i$ in modulus-argument form. **[5]**

18. Prove that $\sin 5\theta = 5\cos^4\theta\sin\theta - 10\cos^2\theta\sin^3\theta + \sin^5\theta$. **[5]**

19. Find the locus of z such that $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$. **[5]**

20. Use de Moivre to evaluate $\int_0^{\pi/2} \cos^4\theta \, \mathrm{d}\theta$. **[5]**

21. Find the fourth roots of -16 and show they form a square on the Argand diagram. **[5]**

22. Solve $z^3 - 3z + 1 = 0$ by relating to $\cos 3\theta$. **[6]**

23. Show that the nth roots of unity sum to zero. **[3]**

24. Find $\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right)^{12}$. **[3]**

25. Express $\sin^6\theta$ in terms of multiple angles. **[5]**

End of Worksheet • *Show all working clearly.*